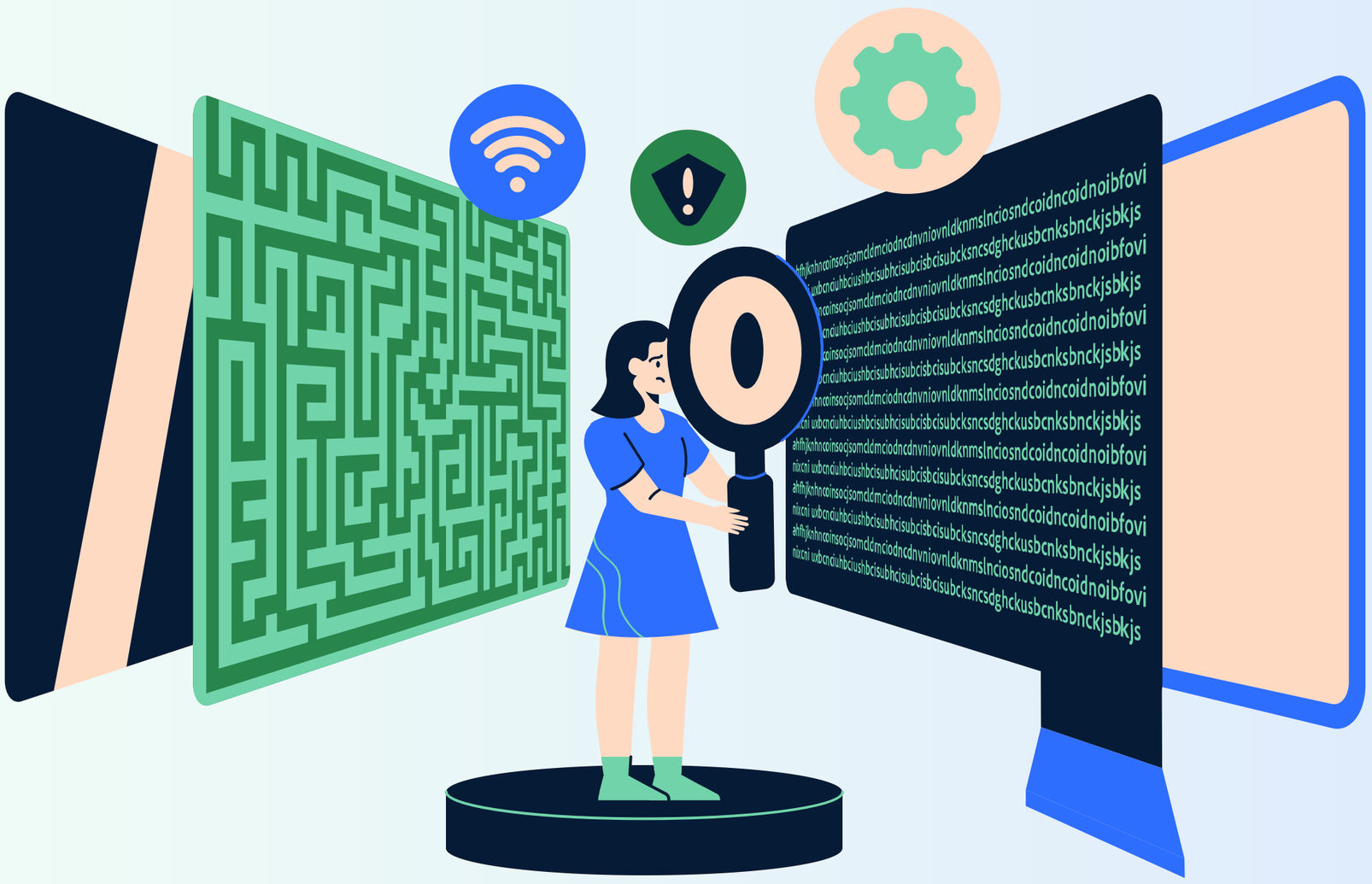
King Abdulaziz University
Faculty of Economics and Administrations
Management Information Systems Department
Business Networks 376
2nd semester 2026

Business Network

Final Project



Section: G
Group:4
Instructor: Dr. Sabah Alsomali

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1. Introduction & Statement of Purpose

Computer networks play an important role in supporting communication, data sharing, and daily business operations.

This project focuses on designing and implementing a Local Area Network (LAN) for a four-story building using Cisco Packet Tracer. Its design will be capable of supporting different departments and each department is separated using VLAN technology to improve network organization, security, and traffic management.

The purpose of this project is to design a secure, organized, and efficient network infrastructure for a multi-floor building environment, which also includes switch configuration, IP addressing, inter-VLAN routing, cable selection, and network security configuration. In addition, connectivity tests are performed to ensure that all devices can communicate successfully within the network.

2. Network Requirements Summary

Floor	Department	Devices	VLAN
1	Lab 1	24PCs	VLAN 2
1	Lab 2	24PCs	VLAN 2
2	Human Resources	12PCs + 1Printer	VLAN 3
3	Marketing	12PCs+1 Printer + 1 Server	VLAN 4
4	IT Department	7PCs + 1 Printer + 1 Server	VLAN 5
4	Management	7PCs + 1 Printer	VLAN 6

3. Network Design & Justification

3.1 Switch Placement

The Core Switch is placed as the main distribution point because it connects the router and all workgroup switches. This design improves communication between floors, reduces cable complexity, and simplifies network management.

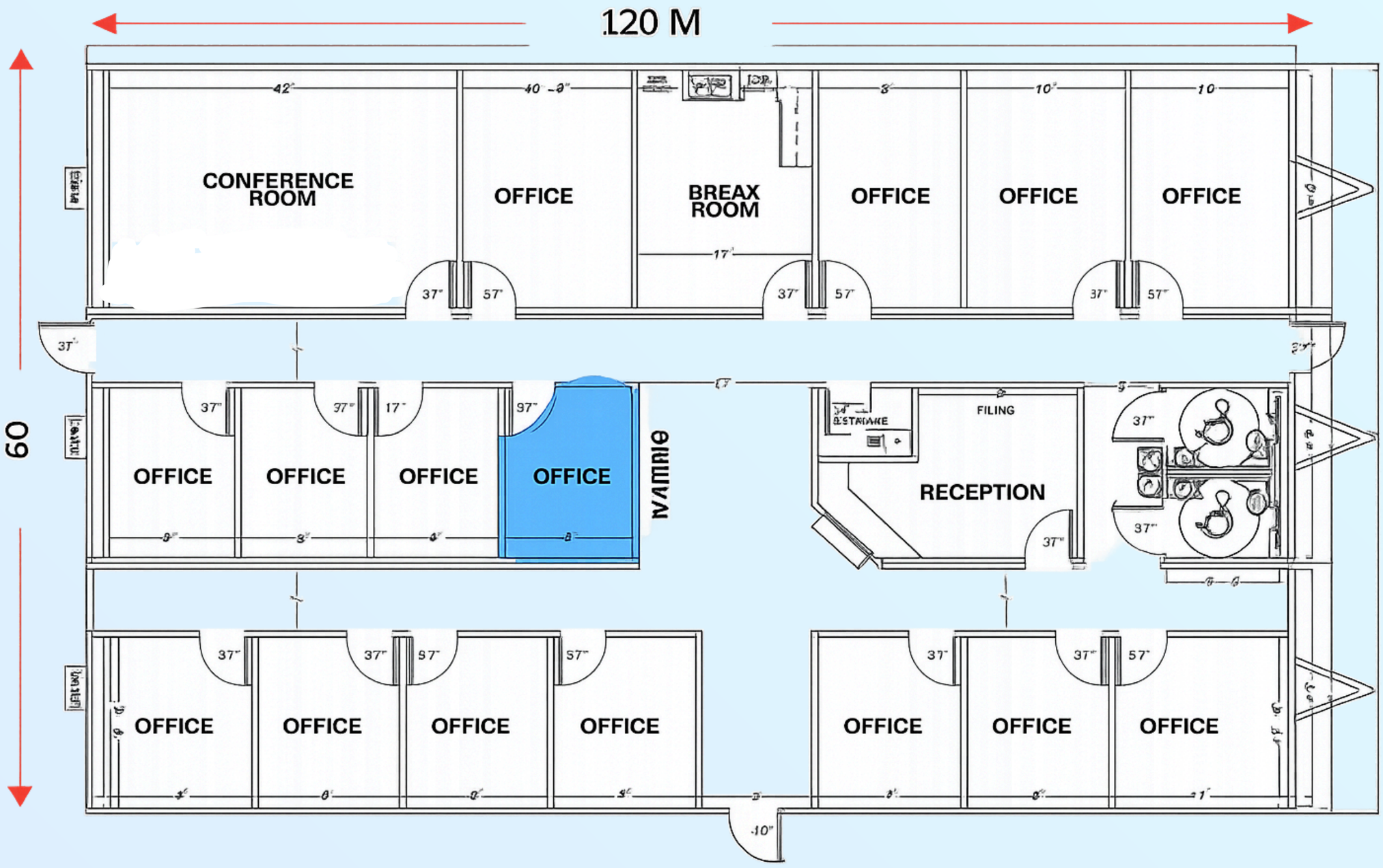
3.2 Rack Cabinet Location

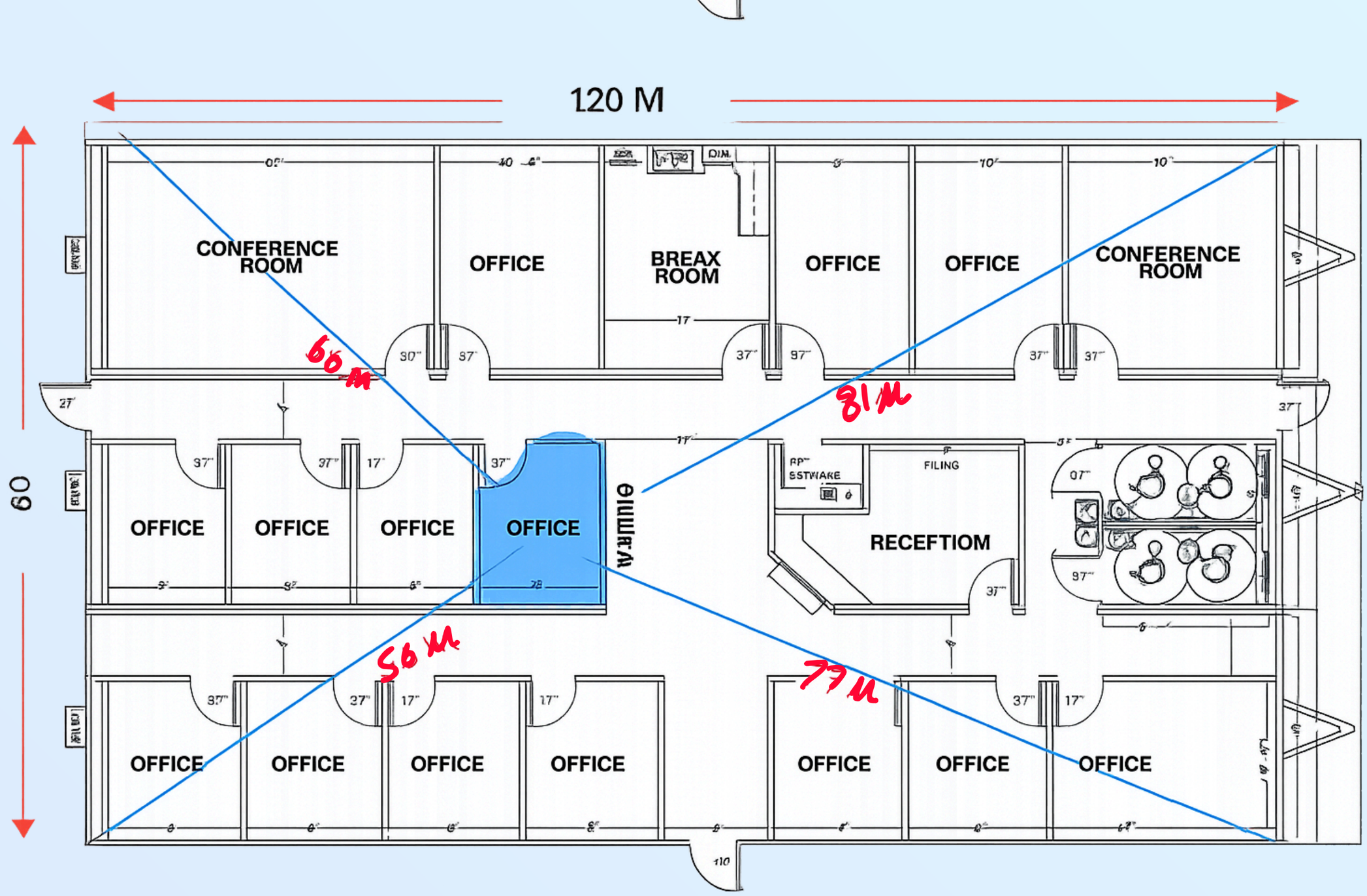
The rack cabinet is in the IT room on the fourth floor. This location is best for network devices, like the core switch, router, and servers. It is safe here.

The IT department is nearby which makes it easier to manage maintain, and troubleshoot the network

This location also helps with cable distribution to all floors. It keeps things organized and efficient. This reduces connection complexity. Improves overall network performance. The rack cabinet and network devices are doing well in this location.

3.2 Rack Cabinet Location





3.3 Cable Selection

Connection	Cable Type	Justification
PC —> Switch	Cat6	Gives 1 Gbps, Afordable
Switch —> Switch	Cat6 UTP	Dependablefor inter-switch and trunking
Server —> Switch	Cat6a or Fiber	Necessary for 10 Gibs bandwidth
Router —> Core Switch	Cat6	Secure connection for external routing

3.4 Switch Management IP Addresses

This table shows the management IP address assigned to each switch. These IP addresses are used to remotely manage and configure the switches in the network.

Table 1: Switch Management IP Addresses

Switch	Management IP
CoreSW	192.168.10.1
Switch0	192.168.10.11
Switch1	192.168.10.12
Switch2	192.168.10.13
Switch3	192.168.10.14

3.5 VLAN Distribution

Due to the large number of hosts on the first floor (48 PCs), two workgroup switches were used to efficiently distribute devices and avoid port limitations.

Table 2: VLAN Distribution

VLAN	Department	Assigned Switch
VLAN2	LAB_166	Switch0 + Switch1
VLAN3	HR	Switch2
VLAN4	Marketing	Switch3
VLAN5	IT	Switch2
VLAN6	Management	Switch3

3.6 VLAN Gateway IP Addresses

Table 3: VLAN Gateway IP Addresses

VLAN	Gateway IP
VLAN2	192.168.15.1
VLAN3	192.168.16.1
VLAN4	192.168.17.1
VLAN5	192.168.18.1
VLAN6	192.168.19.1

We created VLANs for each Department:

```
Core_Switch(config)#do show vlan brief

VLAN Name                                Status    Ports
---- ----------------------------------- --------- -------------------------------
1    default                             active    Gig1/0/6, Gig1/0/7, Gig1/0/8, Gig1/0/9
                                                   Gig1/0/10, Gig1/0/11, Gig1/0/12, Gig1/0/13
                                                   Gig1/0/14, Gig1/0/15, Gig1/0/16, Gig1/0/17
                                                   Gig1/0/18, Gig1/0/19, Gig1/0/20, Gig1/0/21
                                                   Gig1/0/22, Gig1/0/23, Gig1/0/24, Gig1/1/1
                                                   Gig1/1/2, Gig1/1/3, Gig1/1/4
2    LAB_166                             active
3    HR                                  active
4    Marketing                           active
5    IT                                  active
6    Management                          active
1002 fddi-default                        active
1003 token-ring-default                  active
1004 fddinet-default                     active
1005 trnet-default                       active
Core_Switch(config)#
```

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4. Switch Configuration & Security

All switches were configured and secured using the required authentication passwords. The console password was set to **CMIS**, the enable password was set to **EMIS**, and the Telnet password was set to **TMIS**.

Core Switch :

Multilayer Switch0

Physical Config CLI Attributes

IOS Command Line Interface

```
Switch>enable
Switch#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#hostname Core_Switch
Core_Switch(config)#enable secret EMIS
Core_Switch(config)#line console 0
Core_Switch(config-line)#password CMIS
Core_Switch(config-line)#login
Core_Switch(config-line)#exit
Core_Switch(config)#line vty 0 4
Core_Switch(config-line)#password TMIS
Core_Switch(config-line)#login
Core_Switch(config-line)#exit
Core_Switch(config)#service password-encryption
Core_Switch(config)#interface vlan 1
Core_Switch(config-if)#ip address 192.168.10.1 255.255.255.0
Core_Switch(config-if)#no shutdown

Core_Switch(config-if)#
%LINK-3-UPDOWN: Interface Vlan1, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan1, changed state to up

Core_Switch(config-if)#exit
Core_Switch(config)#ip default-gateway 192.168.10.1
Core_Switch(config)#vlan 2
Core_Switch(config-vlan)#name LAB_166
Core_Switch(config-vlan)#exit
Core_Switch(config)#vlan 3
Core_Switch(config-vlan)#name HR
Core_Switch(config-vlan)#exit
Core_Switch(config)#vlan 4
Core_Switch(config-vlan)#name Marketing
Core_Switch(config-vlan)#exit
Core_Switch(config)#vlan 5
Core_Switch(config-vlan)#name IT
Core_Switch(config-vlan)#exit
Core_Switch(config)#vlan 6
Core_Switch(config-vlan)#name Management
Core_Switch(config-vlan)#exit
Core_Switch(config)#
Core_Switch#
%SYS-5-CONFIG_I: Configured from console by console
```

Multilayer Switch0

Physical Config CLI Attributes

IOS Command Line Interface

```
Core_Switch#enable
Core_Switch#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Core_Switch(config)#interface range gig1/0/1-4
Core_Switch(config-if-range)#switchport moode trunk
                            ^
% Invalid input detected at '^' marker.

Core_Switch(config-if-range)#switchport mode trunk



Core_Switch(config-if-range)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet1/0/1, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet1/0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet1/0/2, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet1/0/2, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet1/0/3, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet1/0/3, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet1/0/4, changed state to down

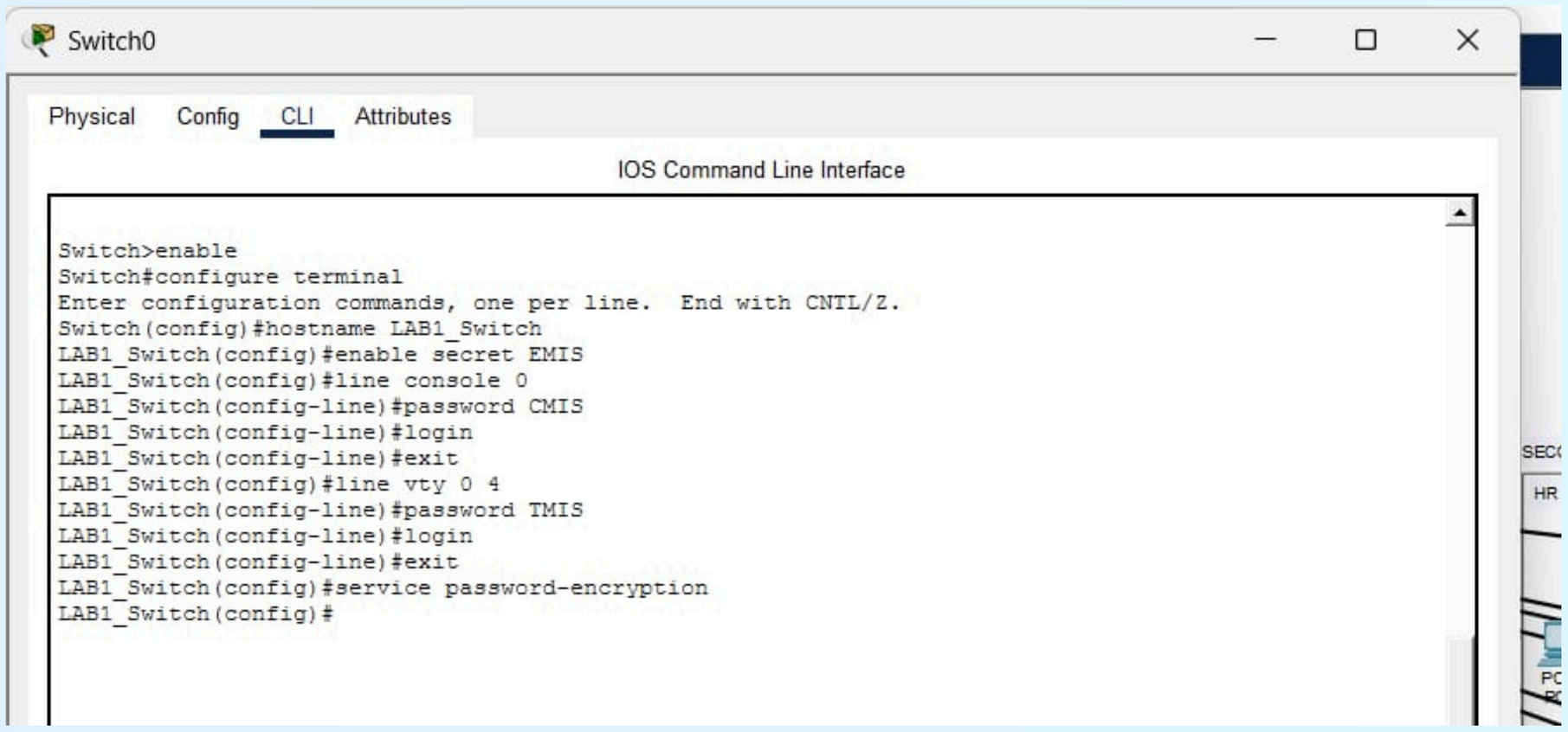
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet1/0/4, changed state to up

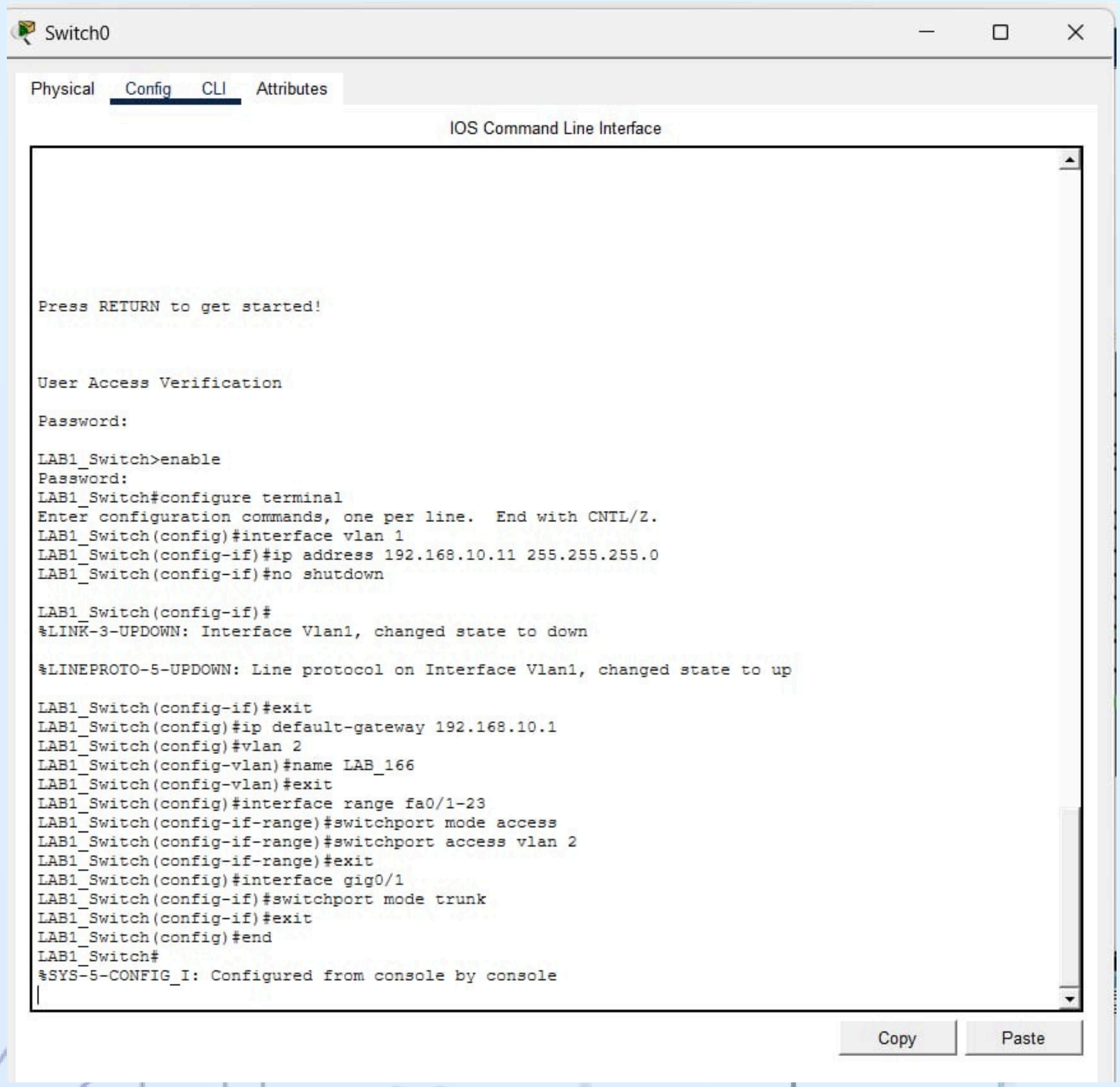
Core_Switch(config-if-range)#exit
Core_Switch(config)#interface gig0/0/0
%Invalid interface type and number
Core_Switch(config)#interface gig0/0/0
%Invalid interface type and number
Core_Switch(config)#interface gig1/0/5
Core_Switch(config-if)#switchport mode trunk
Core_Switch(config-if)#exit
Core_Switch(config)#end
Core_Switch#
%SYS-5-CONFIG_I: Configured from console by console

Core_Switch#copy running-config startup-config
Destination filename [startup-config]?
```

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Switch 0 :


Switch0
Physical
Config
CLI
Attributes
IOS Command Line Interface
Switch>enable
Switch#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#hostname LAB1_Switch
LAB1_Switch(config)#enable secret EMIS
LAB1_Switch(config)#line console 0
LAB1_Switch(config-line)#password CMIS
LAB1_Switch(config-line)#login
LAB1_Switch(config-line)#exit
LAB1_Switch(config)#line vty 0 4
LAB1_Switch(config-line)#password TMIS
LAB1_Switch(config-line)#login
LAB1_Switch(config-line)#exit
LAB1_Switch(config)#service password-encryption
LAB1_Switch(config)#


Switch0
Physical
Config
CLI
Attributes
IOS Command Line Interface
Press RETURN to get started!
User Access Verification
Password:
LAB1_Switch>enable
Password:
LAB1_Switch#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
LAB1_Switch(config)#interface vlan 1
LAB1_Switch(config-if)#ip address 192.168.10.11 255.255.255.0
LAB1_Switch(config-if)#no shutdown
LAB1_Switch(config-if)#
%LINK-3-UPDOWN: Interface Vlan1, changed state to down
%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan1, changed state to up
LAB1_Switch(config-if)#exit
LAB1_Switch(config)#ip default-gateway 192.168.10.1
LAB1_Switch(config)#vlan 2
LAB1_Switch(config-vlan)#name LAB_166
LAB1_Switch(config-vlan)#exit
LAB1_Switch(config)#interface range fa0/1-23
LAB1_Switch(config-if-range)#switchport mode access
LAB1_Switch(config-if-range)#switchport access vlan 2
LAB1_Switch(config-if-range)#exit
LAB1_Switch(config)#interface gig0/1
LAB1_Switch(config-if)#switchport mode trunk
LAB1_Switch(config-if)#exit
LAB1_Switch(config)#end
LAB1_Switch#
%SYS-5-CONFIG_I: Configured from console by console
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Switch 1 :

Switch1

Physical Config CLI Attributes

IOS Command Line Interface

```
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up


Switch>
Switch>enable
Switch#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#hostname LAB2_Switch
LAB2_Switch(config)#enable secret EMIS
LAB2_Switch(config)#line console 0
LAB2_Switch(config-line)#password CMIS
LAB2_Switch(config-line)#login
LAB2_Switch(config-line)#exit
LAB2_Switch(config)#line vty 0 4
LAB2_Switch(config-line)#password TMIS
LAB2_Switch(config-line)#login
LAB2_Switch(config-line)#exit
LAB2_Switch(config)#service password-encryption
LAB2_Switch(config)#interface vlan 1
LAB2_Switch(config-if)#ip address 192.168.10.12 255.255.255.0
LAB2_Switch(config-if)#no shutdown

LAB2_Switch(config-if)#
%LINK-3-UPDOWN: Interface Vlan1, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan1, changed state to up

LAB2_Switch(config-if)#exit
LAB2_Switch(config)#ip default-gateway 192.168.10.1
LAB2_Switch(config)#vlan 2
LAB2_Switch(config-vlan)#name LAB_166
LAB2_Switch(config-vlan)#exit
LAB2_Switch(config)#interface range fa0/1-23
LAB2_Switch(config-if-range)#switchport mode access
LAB2_Switch(config-if-range)#switchport access vlan 2
LAB2_Switch(config-if-range)#exit
LAB2_Switch(config)#interface gig0/1
LAB2_Switch(config-if)#switchport mode trunk
LAB2_Switch(config-if)#exit
LAB2_Switch(config)#end
LAB2_Switch#
%SYS-5-CONFIG_I: Configured from console by console
```

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Top

Switch 2 :

Switch2

Physical Config CLI Attributes

IOS Command Line Interface

```
Switch>enable
Switch#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#hostname IT_HR_SW
IT_HR_SW(config)#enable secret EMIS
IT_HR_SW(config)#line console 0
IT_HR_SW(config-line)#password CMIS
IT_HR_SW(config-line)#login
IT_HR_SW(config-line)#exit
IT_HR_SW(config)#line vty 0 4
IT_HR_SW(config-line)#password TMIS
IT_HR_SW(config-line)#login
IT_HR_SW(config-line)#exit
IT_HR_SW(config)#service password-encryption
IT_HR_SW(config)#interface vlan 1
IT_HR_SW(config-if)#ip address 192.168.10.13 255.255.255.0
IT_HR_SW(config-if)#no shutdown

IT_HR_SW(config-if)#
%LINK-3-UPDOWN: Interface Vlan1, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan1, changed state to up

IT_HR_SW(config-if)#exit
IT_HR_SW(config)#ip default-gateway 192.168.10.1
IT_HR_SW(config)#vlan 3
IT_HR_SW(config-vlan)#name HR
IT_HR_SW(config-vlan)#exit
IT_HR_SW(config)#vlan 5
IT_HR_SW(config-vlan)#name IT
IT_HR_SW(config-vlan)#exit
IT_HR_SW(config)#interface range fa0/9-21
IT_HR_SW(config-if-range)#switchport mode access
IT_HR_SW(config-if-range)#switchport access vlan 3
IT_HR_SW(config-if-range)#exit
IT_HR_SW(config)#interface range fa0/1-8
IT_HR_SW(config-if-range)#switchport mode access
IT_HR_SW(config-if-range)#switchport access vlan 5
IT_HR_SW(config-if-range)#exit
IT_HR_SW(config)#interface gig0/2
IT_HR_SW(config-if)#switchport mode trunk
IT_HR_SW(config-if)#exit
IT_HR_SW(config)#end
IT_HR_SW#
%SYS-5-CONFIG_I: Configured from console by console
```

Copy Paste

switch 3 :

Switch3

Physical | Config | CLI | Attributes

IOS Command Line Interface

```
Switch>enable
Switch#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#hostname MG_MKT_SW
MG_MKT_SW(config)#enable secret EMIS
MG_MKT_SW(config)#line console 0
MG_MKT_SW(config-line)#password CMIS
MG_MKT_SW(config-line)#login
MG_MKT_SW(config-line)#exit
MG_MKT_SW(config)#line vty 0 4
MG_MKT_SW(config-line)#password TMIS
MG_MKT_SW(config-line)#login
MG_MKT_SW(config-line)#exit
MG_MKT_SW(config)#service password-encryption
MG_MKT_SW(config)#interface vlan 1
MG_MKT_SW(config-if)#ip address 192.168.10.14 255.255.255.0
MG_MKT_SW(config-if)#no shutdown

MG_MKT_SW(config-if)#
%LINK-3-UPDOWN: Interface Vlan1, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan1, changed state to up

MG_MKT_SW(config-if)#exit
MG_MKT_SW(config)#ip default-gateway 192.168.10.1
MG_MKT_SW(config)#vlan 4
MG_MKT_SW(config-vlan)#name Marketing
MG_MKT_SW(config-vlan)#exit
MG_MKT_SW(config)#vlan 6
MG_MKT_SW(config-vlan)#name Management
MG_MKT_SW(config-vlan)#exit
MG_MKT_SW(config)#interface range fa0/9-22
MG_MKT_SW(config-if-range)#switchport mode access
MG_MKT_SW(config-if-range)#switchport access vlan 4
MG_MKT_SW(config-if-range)#exit
MG_MKT_SW(config)#interface range fa0/1-8
MG_MKT_SW(config-if-range)#switchport mode access
MG_MKT_SW(config-if-range)#switchport access vlan 6
MG_MKT_SW(config-if-range)#exit
MG_MKT_SW(config)#interface gig0/2
MG_MKT_SW(config-if)#switchport mode trunk
MG_MKT_SW(config-if)#exit
MG_MKT_SW(config)#end
MG_MKT_SW#
%SYS-5-CONFIG_I: Configured from console by console
```

Copy | Paste

Top

Router:

Router0

Physical Config CLI Attributes

IOS Command Line Interface

```
Router>enable
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#interface gig0/0/0
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0/0, changed state to up

Router(config-if)#exit
Router(config)#interface gig0/0/0.2
Router(config-subif)#
%LINK-3-UPDOWN: Interface GigabitEthernet0/0/0.2, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0/0.2, changed state to up

Router(config-subif)#encapsulation dot1Q 3
Router(config-subif)#no encapsulation dot1Q 3
Router(config-subif)#exit
Router(config)#interface gig0/0/0.2
Router(config-subif)#encapsulation dot1Q 2
Router(config-subif)#ip address 192.168.15.1 255.255.255.0
Router(config-subif)#exit
Router(config)#interface gig0/0/0.3
Router(config-subif)#
%LINK-3-UPDOWN: Interface GigabitEthernet0/0/0.3, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0/0.3, changed state to up

Router(config-subif)#encapsulation dot1Q 3
Router(config-subif)#ip address 192.168.16.1 255.255.255.0
Router(config-subif)#exit
Router(config)#interface gig0/0/0.4
Router(config-subif)#
%LINK-3-UPDOWN: Interface GigabitEthernet0/0/0.4, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0/0.4, changed state to up

Router(config-subif)#encapsulation dot1Q 4
Router(config-subif)#ip address 192.168.17.1 255.255.255.0
Router(config-subif)#exit
```

Router0

Physical Config CLI Attributes

IOS Command Line Interface

```
Router(config-subif)#exit
Router(config)#interface gig0/0/0.2
Router(config-subif)#encapsulation dot1Q 2
Router(config-subif)#ip address 192.168.15.1 255.255.255.0
Router(config-subif)#exit
Router(config)#interface gig0/0/0.3
Router(config-subif)#
%LINK-3-UPDOWN: Interface GigabitEthernet0/0/0.3, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0/0.3, changed state to up

Router(config-subif)#encapsulation dot1Q 3
Router(config-subif)#ip address 192.168.16.1 255.255.255.0
Router(config-subif)#exit
Router(config)#interface gig0/0/0.4
Router(config-subif)#
%LINK-3-UPDOWN: Interface GigabitEthernet0/0/0.4, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0/0.4, changed state to up

Router(config-subif)#encapsulation dot1Q 4
Router(config-subif)#ip address 192.168.17.1 255.255.255.0
Router(config-subif)#exit
Router(config)#interface gig0/0/0.5
Router(config-subif)#
%LINK-3-UPDOWN: Interface GigabitEthernet0/0/0.5, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0/0.5, changed state to up

Router(config-subif)#encapsulation dot1Q 5
Router(config-subif)#ip address 192.168.18.1 255.255.255.0
Router(config-subif)#exit
Router(config)#interface gig0/0/0.6
Router(config-subif)#
%LINK-3-UPDOWN: Interface GigabitEthernet0/0/0.6, changed state to down

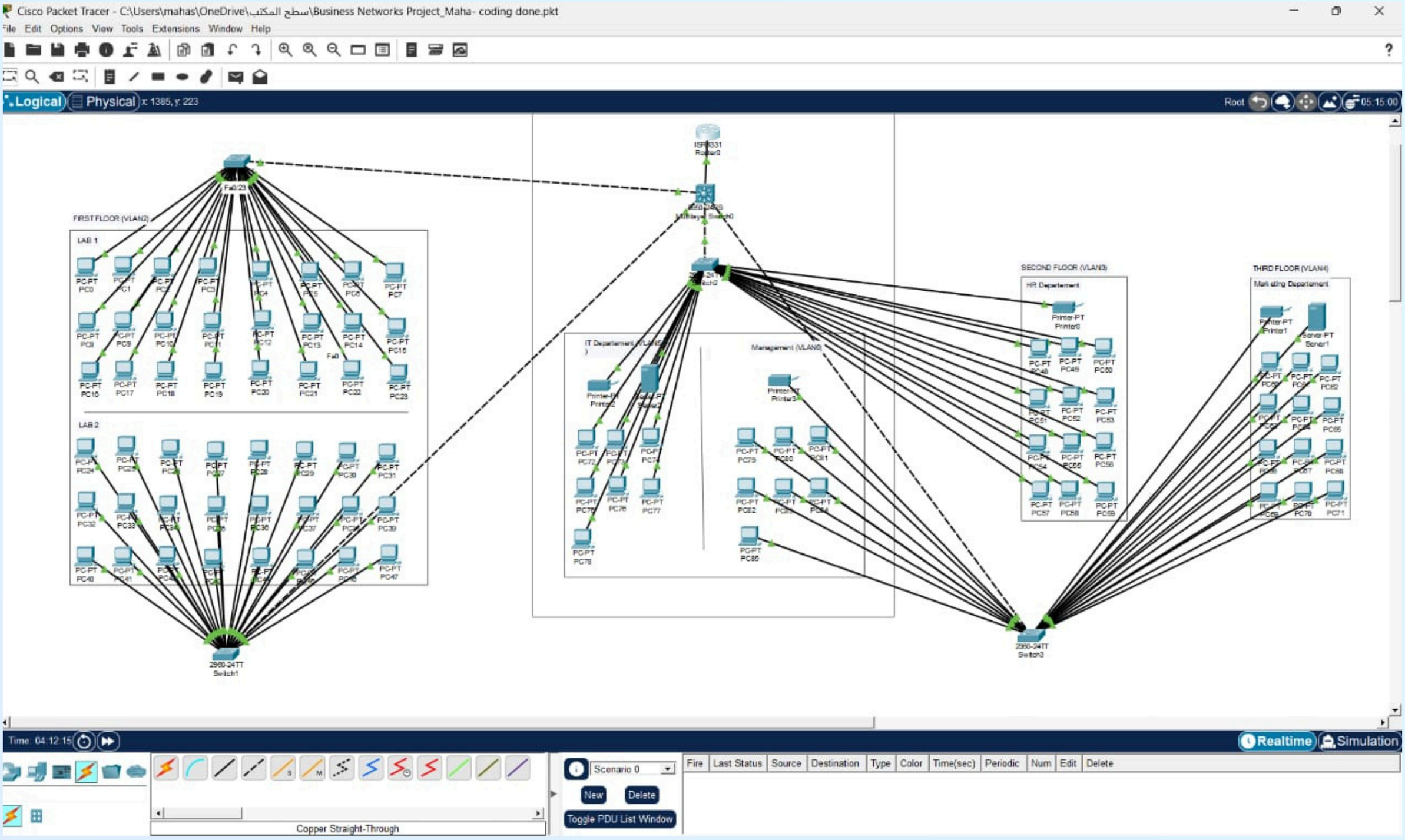
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0/0.6, changed state to up

Router(config-subif)#encapsulation dot1Q 6
Router(config-subif)#ip address 192.168.19.1 255.255.255.0
Router(config-subif)#exit
Router(config)#end
Router#
%SYS-5-CONFIG_I: Configured from console by console
```

Copy Paste

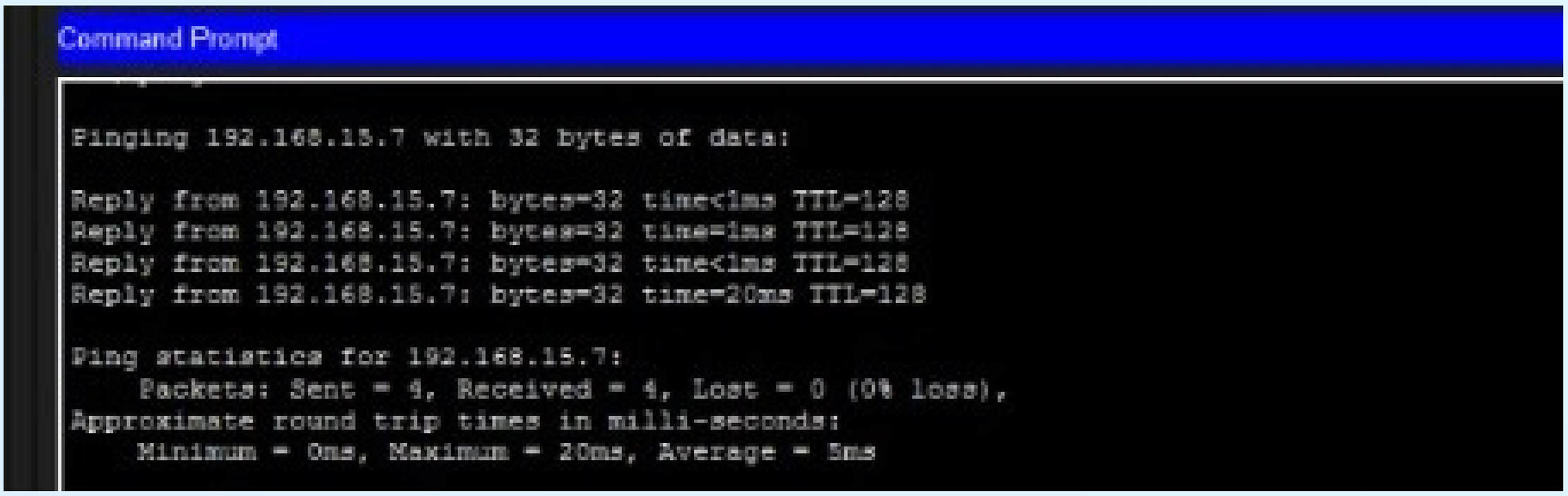
5. **Company Network Structure**

This network connects all departments and floors using switches, PC’s, printers, and VLANs for better communication and management.

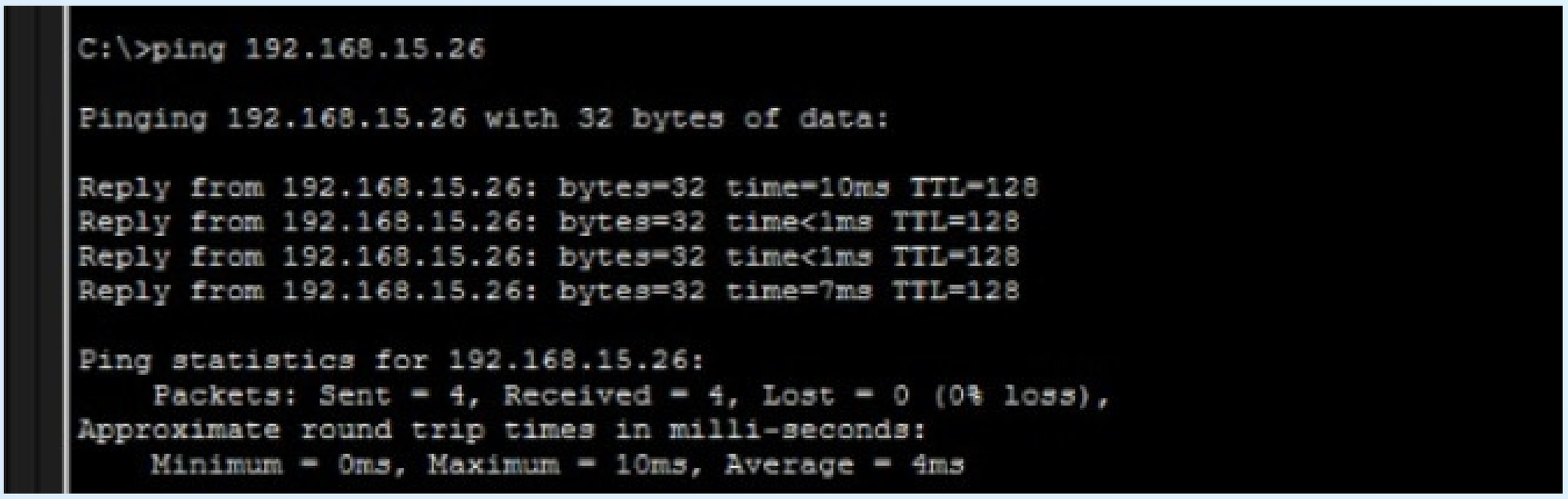


6. **Verification & Testing**

First, we verified communication between devices in the same lab and the same VLAN (VLAN2 – LAB_166).



Next, we verified communication between devices connected to the same switch between lab 1 and lab .



6. **Verification & Testing**

Then, we verified communication between VLAN2 and VLAN3.

```
C:\>ping 192.168.16.2

Pinging 192.168.16.2 with 32 bytes of data:

Request timed out.
Reply from 192.168.16.2: bytes=32 time<1ms TTL=127
Reply from 192.168.16.2: bytes=32 time=11ms TTL=127
Reply from 192.168.16.2: bytes=32 time<1ms TTL=127

Ping statistics for 192.168.16.2:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 11ms, Average = 3ms
```

After that, we verified communication between VLAN2 and VLAN4

```
C:\>ping 192.168.17.2

Pinging 192.168.17.2 with 32 bytes of data:

Request timed out.
Reply from 192.168.17.2: bytes=32 time=10ms TTL=127
Reply from 192.168.17.2: bytes=32 time=3ms TTL=127
Reply from 192.168.17.2: bytes=32 time=3ms TTL=127

Ping statistics for 192.168.17.2:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
Approximate round trip times in milli-seconds:
    Minimum = 3ms, Maximum = 10ms, Average = 5ms

C:\>
```

6. **Verification & Testing**

Next, we verified communication between VLAN2 and VLAN5.

```
C:\>ping 192.168.18.2

Pinging 192.168.18.2 with 32 bytes of data:

Reply from 192.168.18.2: bytes=32 time=2ms TTL=127
Reply from 192.168.18.2: bytes=32 time<1ms TTL=127
Reply from 192.168.18.2: bytes=32 time<1ms TTL=127
Reply from 192.168.18.2: bytes=32 time<1ms TTL=127

Ping statistics for 192.168.18.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 2ms, Average = 0ms
```

Finally, we verified communication between VLAN2 and VLAN6.

```
C:\>ping 192.168.19.2

Pinging 192.168.19.2 with 32 bytes of data:

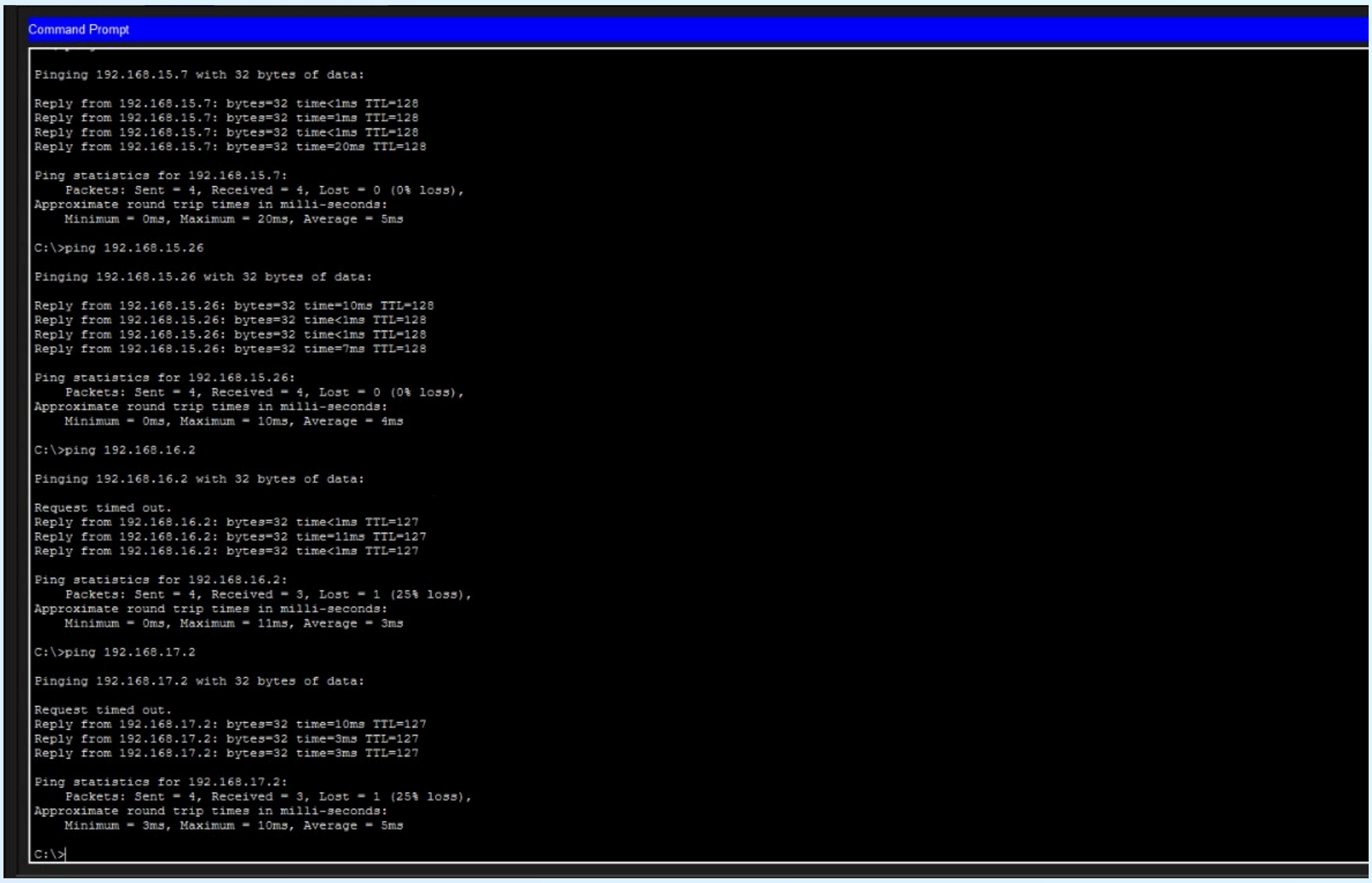
Reply from 192.168.19.2: bytes=32 time<1ms TTL=127
Reply from 192.168.19.2: bytes=32 time=3ms TTL=127
Reply from 192.168.19.2: bytes=32 time=11ms TTL=127
Reply from 192.168.19.2: bytes=32 time<1ms TTL=127

Ping statistics for 192.168.19.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 11ms, Average = 3ms

C:\>
```

6. **Verification & Testing**

The following ping results from PC6 in LAB1 were used to verify successful communication between the same and the different VLANs in the network.



7. Challenges, Solutions & Critical Thinking

1. **High Host Density on the First Floor**: There are a many computers on the first floor 48 PCs, this is a challenge. To deal with this and prevent running out of ports, the network uses two workgroup switches to connect all the devices.

2. **Cable Complexity in a Multi-Floor Building**: When you have to connect devices on four floors it can get really messy with all the cables. To solve this problem the core switch is used as the connection point and the rack cabinet is placed in the IT room on the 4th floor. This makes the connections less complicated. Improves the overall network performance.

3. **Departmental Traffic Management and Security**: The network has to support different departments like HR, Marketing and IT. This means we need to organize the traffic flow and protect the data. The network design uses VLAN technology to separate the departments, which makes the network more organized and secure.

4. **High Bandwidth Requirements for Servers**: The servers need many bandwidth to work properly and standard cables might not be able to handle it. To solve this, the network uses Cat6a or Fiber cables to connect the servers to the switches, which gives them the 10 Gbps bandwidth they require.

5. **Secure Remote Management**: Managing the network from a distance can be a security challenge. To solve this problem each switch is given a management IP address and uses layered authentication passwords, like Console, Enable and Telnet to make sure only authorized IT personnel can access the hardware. The network design uses these solutions to deal with the challenges and make the network work better.

8. **Conclusion**

In conclusion, this project successfully designed a secure, organized and efficient network infrastructure for a business that is spread out over many floors.

The network was made to help different departments efficiently through the use of VLANs, proper IP addressing, and inter VLAN routing. Also, suitable cable types, switch placement, and security configurations were selected to improve network performance, make management easier, and ensure reliable communication between devices.

We put the rack cabinet in the 4th-floor IT room and used the Core Switch as a central distribution point which really helped cut down on the number of cables and made it easier to fix problems when they came up. When we had some tough technical problems like having to deal with a lot of computers on the first floor our plan made sure that every device could still get online without any issues. So by choosing the cables and using strong security measures we made a secure organized and efficient network infrastructure that can support the business and help it run smoothly every day.